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DEPARTMENT OF CSE(DATA SCIENCE)

A Project Report On

"CRIMINAL ACTIVITY MONITORING SYSTEM"

A report submitted in partial fulfillment of the requirements for the

DATA VISUALIZATION WITH POWER BI

Submitted By

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Belagavi, Karnataka 2025-2026

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CERTIFICATE

This is to certify that the open ended project report on DATA VISUALIZATION WITH POWER BI **"Criminal Activity Monitoring System"** has been successfully presented by **MAHAMMED JUNED M** bearing USN **3BR24CD402** student of semester B.E for the partial fulfillment of the requirements for the award of **Bachelor Degree in CSE(DS)** of the BALLARI INSTITUTE OF TECHNOLOGY& MANAGEMENT, BALLARI during the

academic year 2025-2026.

Signature of guide	Signature of HOD
Mr. Sundeep Sindhanoor	**Dr. Aradhana D**

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CRIMINAL ACTIVITY MONITORING SYSTEM

Abstract

This report presents a comprehensive analysis of crime data across major Indian cities using a Power BI interactive dashboard. The dataset covers reported crimes spanning multiple cities, crime domains, and years (2020–2024). The objective is to derive actionable insights from crime data to support law enforcement agencies, policy makers, and urban planners in making evidence-based decisions. Key metrics analyzed include Total Reported Crimes (14K), Pending Cases (7,261), Closed Cases (7,172), and an overall Closure Rate of 0.50 (50%). The dashboard categorizes crimes into four domains: Other Crime (57.13%), Violent Crime (26.82%), Fire Accident (6.61%), and Traffic Fatality (9.44%).

Chapter 1: Introduction

1.1 Background

Crime analysis is a critical function of modern law enforcement and urban governance. With increasing urbanization, Indian cities are witnessing diverse crime patterns across different geographic zones. Systematic analysis of crime data helps authorities allocate resources effectively, identify crime hotspots, evaluate closure efficiency, and design preventive strategies. This project leverages Power BI to visualize and analyze crime records from 30+ Indian cities over a five-year period (2020–2024), covering four crime domains: Other Crime, Violent Crime, Fire Accident, and Traffic Fatality.

1.2 Problem Statement

Despite vast amounts of crime data being collected by police departments across Indian cities, the absence of a unified, interactive analytical framework makes it difficult to identify trends, compare city-level performance, or measure case closure efficiency. Without such visibility, law enforcement agencies risk making decisions based on incomplete or outdated information, leading to suboptimal resource deployment and delayed justice delivery.

1.3 Objectives

- Analyze total reported crimes, pending cases, closed cases, and closure rates across major Indian cities.
- Identify the top 10 cities with the highest crime counts and their case management efficiency.
- Evaluate crime distribution by domain (Other Crime, Violent Crime, Fire Accident, Traffic Fatality).
- Track monthly crime trends to identify seasonal patterns and peak crime periods.
- Visualize city-wise and year-wise crime trends using interactive maps, bar charts, and line graphs.

1.4 Scope

The dashboard is designed for use by law enforcement agencies, government policy departments, urban safety planners, and academic researchers to monitor city-level crime performance, benchmark case closure rates, and guide strategic deployment of police resources.

Chapter 2: Data Description

Problem Statement

The dataset underpinning this dashboard encompasses city-level crime records with the following key attributes:

- City: Name of the Indian city (30+ cities covered including Delhi, Mumbai, Bangalore, Hyderabad, Kolkata, Chennai, Pune, etc.)
- Crime Domain: Category of crime — Other Crime, Violent Crime, Fire Accident, Traffic Fatality
- Year: Year of crime occurrence (2020, 2021, 2022, 2023, 2024)
- Month: Month of reporting (January through December)
- Total Crimes: Number of crimes reported
- Open Cases: Number of cases still under investigation
- Closed Cases: Number of cases resolved or closed
- Closure Rate %: Ratio of closed cases to total reported crimes

The data spans five years (2020–2024) across 30+ Indian cities, covering four crime domains with monthly granularity, enabling both temporal trend analysis and city-level benchmarking.

Objectives

The primary objective of this project is to design and deploy an interactive Power BI dashboard that enables multi-dimensional crime data analysis. The dashboard aims to empower stakeholders with actionable insights for improving policing strategies, resource allocation, and public safety outcomes across Indian cities.

Chapter 3: System Requirements

Functional Requirements

- Display real-time KPI metrics: Total Reported Crimes, Pending Cases, Closed Cases, Closure Rate %.
- Provide interactive filtering by City, Crime Domain, and Year.
- Visualize geographic crime distribution using an interactive map with pie charts per city.
- Show monthly crime trend analysis using line charts.
- Present top 10 cities by crime volume using bar charts.
- Display crime domain distribution using donut charts.

Non-Functional Requirements

- Performance: Dashboard must render within 3 seconds for complete dataset of 14K+ records.
- Usability: Intuitive navigation with color-coded crime domain indicators.
- Scalability: System should support addition of new cities and years without structural redesign.
- Accuracy: All KPI values must be validated against raw source data with zero tolerance for calculation errors.

Hardware Requirements

- Processor: Intel Core i5 or equivalent (minimum)
- RAM: 8 GB minimum (16 GB recommended for large datasets)
- Storage: 500 GB HDD / 256 GB SSD minimum
- Display: 1920x1080 resolution or higher for optimal dashboard viewing

Software Requirements

- Power BI Desktop: For dashboard development, data modelling, and interactive visualization.
- Microsoft Excel / CSV: For raw data storage and initial preprocessing.
- DAX (Data Analysis Expressions): For creating calculated measures such as Total Crimes, Closed Cases, Pending Cases, and Closure Rate %.
- Power Query: For data transformation, cleaning, and shaping prior to loading.
- Operating System: Windows 10 / Windows 11 (64-bit)

Chapter 4: Methodology

Data Collection

Crime data was collected from publicly available datasets covering Indian cities. The dataset includes city-level crime records categorized by crime domain, year, and month. Data integrity checks were performed to ensure completeness and consistency across all records before loading into Power BI.

Data Preparation (Power Query)

Power Query was used to clean and transform the raw data. Steps included: removing null and duplicate records, standardizing city names and crime domain labels, creating derived columns for Month Number and Year, and validating numeric fields (Total Crimes, Open Cases, Closed Cases). The cleaned dataset was then loaded into the Power BI data model.

Data Visualization

Multiple visualization types were employed to represent different facets of the crime data: KPI cards for top-line metrics, a Bing map with pie chart overlays for geographic crime distribution, bar charts for top 10 city comparisons, a donut chart for crime domain distribution, a line chart for monthly trends, and a multi-line chart for year-wise city comparisons.

Filters and Interactivity

Slicers were implemented for Crime Domain (Fire Accident, Other Crime, Traffic Fatality, Violent Crime) allowing users to filter all dashboard visuals simultaneously. Cross-filtering between visuals enables drill-down from city-level to crime-domain-level analysis in a single click.

Dashboard Design

The dashboard consists of two pages: (1) Crime Data Analysis — the primary summary page featuring KPI cards, monthly trend line, crime domain donut chart, and city-wise bar chart; and (2) Geographic Analysis — featuring an interactive Bing map with pie overlays, top 10 cities bar chart, city-level closure rate table, and year-wise city trend lines.

Outcome

The resulting dashboard provides a holistic, interactive view of crime patterns across India, enabling stakeholders to quickly identify cities with high crime volumes, monitor closure rate efficiency, and track year-over-year crime trends for each city and crime domain.

Chapter 5: Implementation

Dataset Description

The dataset contains crime records for 30+ Indian cities over five years (2020–2024). Each row represents monthly crime statistics for a specific city and crime domain combination. The dataset includes fields: City, Crime Domain, Year, Month, Total Crimes, Open Cases, Closed Cases, and Closure Rate %.

Data Loading

The CSV dataset was imported into Power BI Desktop using the "Get Data" option. Data types were verified and corrected — numeric columns (Total Crimes, Open Cases, Closed Cases) were set to Whole Number, and date-related fields were set to Text for controlled sorting via a Month Order column.

Data Transformation (Power Query)

In Power Query Editor: removed blank rows and duplicate records, standardized city name formatting (title case), validated that Closed Cases + Open Cases = Total Crimes for each row, added a Month Order (1–12) column for correct chronological sorting in line charts, and trimmed whitespace from Crime Domain column values.

Data Modelling

A flat single-table data model was used given the nature of the dataset. All relationships and aggregations were handled via DAX measures. A separate Date/Month ordering table was created and linked via Month Order to ensure correct chronological display in time-series visualizations.

DAX Measures (Key Calculations)

- Total Crimes: SUM(CrimeData[Total Crimes])
- Closed Cases: SUM(CrimeData[Closed Cases])
- Pending Cases: [Total Crimes] – [Closed Cases]
- Closure Rate %: DIVIDE([Closed Cases], [Total Crimes], 0)
- Crime Domain % Share: DIVIDE(CALCULATE([Total Crimes], ALLEXCEPT(CrimeData, CrimeData[Crime Domain])), CALCULATE([Total Crimes], ALL(CrimeData[Crime Domain])), 0)

Dashboard Visualization

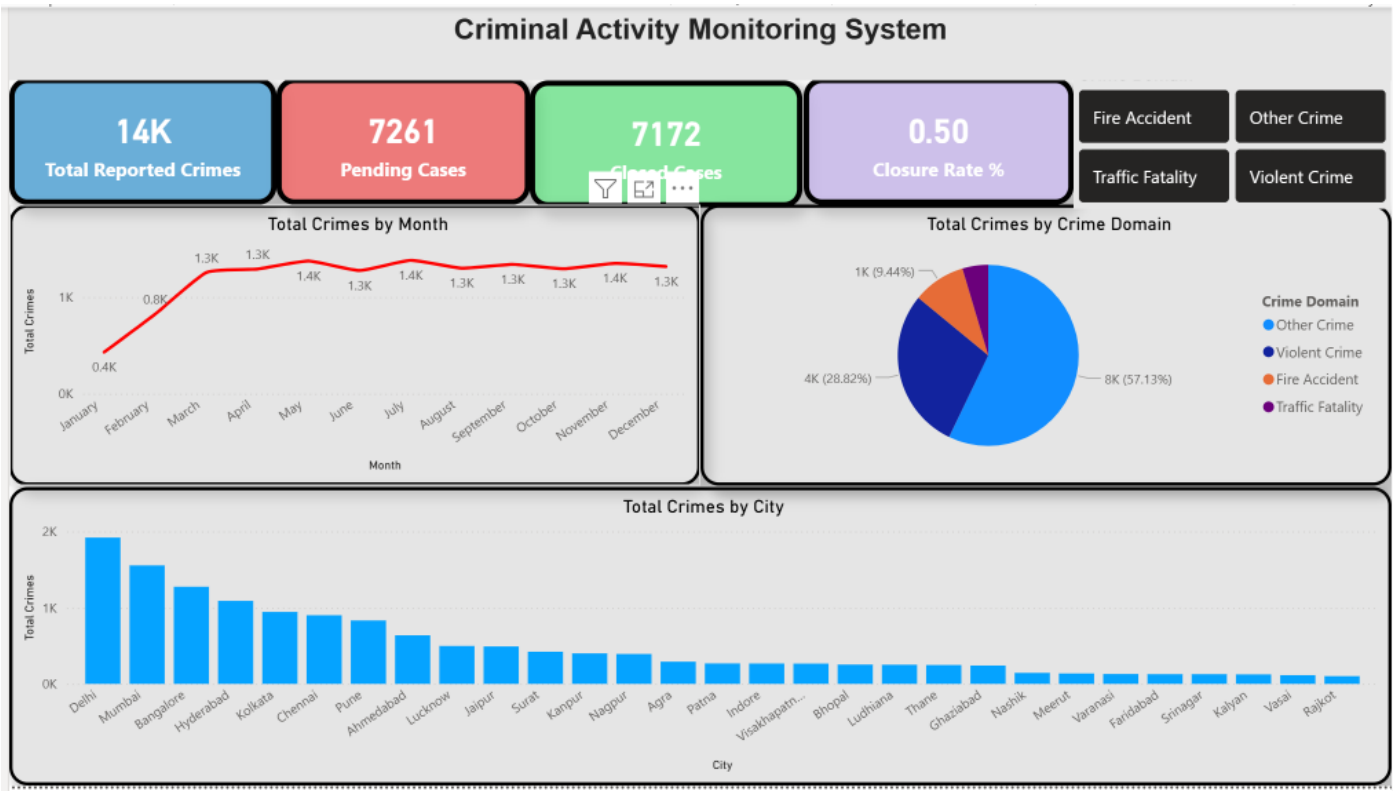
- KPI Cards: Four colored tiles displaying Total Reported Crimes (14K — blue), Pending Cases (7,261 — red), Closed Cases (7,172 — green), Closure Rate % (0.50 — purple).
- Monthly Trend Chart: Line chart showing Total Crimes by Month, peaking between April–October (~1.3K–1.4K crimes/month) and dipping in January (0.4K).
- Crime Domain Donut Chart: Displays domain share: Other Crime 57.13% (8K), Violent Crime 26.82% (4K), Traffic Fatality 9.44% (1K), Fire Accident 6.61%.
- City-wise Bar Chart: Horizontal bar chart of Total Crimes by City for all 30+ cities, with Delhi leading.
- Geographic Map: Interactive Bing map with pie chart bubbles at each city location, colored by crime domain, enabling geographic drill-down.

Chapter 6: Result

Top 10 Cities by Total Crimes

The bar chart on the geographic dashboard identifies the ten cities with the highest total crime counts. Delhi leads significantly, followed by Mumbai and Bangalore:

Rank	City	Total Crimes
1	Delhi	1,919
2	Mumbai	1,554
3	Bangalore	1,274
4	Hyderabad	1,088
5	Kolkata	943
6	Chennai	900
7	Pune	832
8	Ahmedabad	637
9	Lucknow	497
10	Jaipur	491



Conclusion

The Crime Data Analysis Dashboard represents an effective data visualization solution built on Power BI that transforms raw crime records into actionable intelligence. By integrating city-level crime metrics, crime domain categorizations, monthly and annual trend analysis, and geographic visualization, the dashboard empowers law enforcement agencies and policy makers with clear, evidence-based insights.

Key findings reveal that Delhi, Mumbai, and Bangalore are the highest-crime cities requiring priority attention. Other Crime dominates at 57.13% of all incidents, suggesting the need for community policing and preventive measures. The overall 50% closure rate indicates room for significant improvement in investigation and case resolution capabilities. Monthly trends confirm that crime activity peaks from March through November, enabling targeted resource deployment during high-risk periods. Delhi's declining trend in 2024 offers an encouraging data point that targeted interventions can yield measurable results.

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